Problem Landscape and Success Criteria for One-User Offline-to-Online Personalized Policies

Research Position and Evaluation Contract

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Abstract

We study one persistent policy for one user. The policy is initialized from that user's thoughts, conversations, corrections, and task trajectories, then updated from later interactions. This setting joins problems that the literature usually studies separately: sparse preference identification, offline initialization, online adaptation, continual learning, agent credit assignment, user simulation, privacy, and efficient deployment. We organize these problems, state the claim that best distinguishes our work, and define the evidence required to support it. The central test is not whether a model can recall a profile. It is whether a user-owned policy can improve on untouched future tasks after real parameter updates, while task success and safety remain non-inferior.

1 Setting

Evidence base. This synthesis uses both project PRISMA runs: 6,663 raw paper records before cross-run deduplication, plus 119 stored full-text review artifacts. We use the broad corpus to identify recurring problems, then cite representative primary papers for each problem. A citation below is evidence for that paper's method or evaluation target, not evidence that our proposed policy already works.

One user is the unit of personalization. For user u, the offline history

$$H_u^{(0)} = \{\text{thoughts, conversations, corrections, preferences, task states, actions, tools, and outcomes}\}$$

produces an initial checkpoint $c_u^{(0)}$. After deployment, interaction batch $D_u^{(t)}$ is generated by the active checkpoint $c_u^{(t)}$. An update algorithm $\mathcal{A}$ emits

$$c_u^{(t+1)} = \mathcal{A}\Big(c_u^{(t)}, H_u^{(0)}, D_u^{(1:t)}\Big). \tag{1}$$

Every record names its user, time, task, generating checkpoint, action trace, feedback source, and training eligibility. A shared backbone or population prior is allowed, but another user's raw history cannot silently enter user u's deployed branch.

A policy update is required. Retrieval, profiles, memory, and decoding control are strong baselines, not the learned object. OPPU stores one user's history in a private PEFT module [Tan et al., 2024]; P-RLHF learns user-conditioned policy parameters from preference pairs [Li et al., 2024]; and SPRInG updates one user adapter over chronological periods [Kim and Kim, 2026]. These are the closest parameter-learning precedents. CoSteer instead performs tuning-free local delta steering at decoding time [Lv et al., 2025], while FABLE updates a small external execution policy around a frozen host agent [Jin et al., 2026]. Both are important deployment baselines, but neither is the same claim as improving the user's language-model policy weights.

The objective is constrained, not a single blended score. The method should improve future personalization while preserving task competence and safety:

$$\begin{aligned}\max_{\mathcal{A}} \quad & \mathbb{E}[U_{\text{pers}}(c_u^{(t+1)})] \\ \text{s.t.} \quad & \mathbb{E}[U_{\text{task}}(c_u^{(t+1)}) - U_{\text{task}}(c_u^{(t)})] \geq -\delta_{\text{task}}, \\ & \mathbb{E}[U_{\text{safe}}(c_u^{(t+1)}) - U_{\text{safe}}(c_u^{(t)})] \geq -\delta_{\text{safe}}, \\ & C_{\text{train}}, C_{\text{serve}}, C_{\text{feedback}} \text{ remain within declared budgets.}\end{aligned} \tag{2}$$

Task success and personalization are therefore co-primary measurements. One cannot compensate numerically for the failure of the other.

2 Problems Studied in This Setting

2.1 Learning the user

P1: identify preferences from sparse, mixed evidence. Aggregating feedback across users can erase minority or conflicting preferences [Li et al., 2024, Jang et al., 2023]. A single user also supplies heterogeneous evidence: explicit requests, implicit choices, edits, accepted outputs, and task outcomes do not share one label space. CoPL and PROPER borrow structure from related users to mitigate sparse histories [Choi et al., 2025, Zhang et al., 2025]; FSPO studies rapid adaptation from few preference examples [Singh et al., 2025]. The open question for us is whether one policy can use the target user's complete history without reducing it to a textual profile.

P2: separate persistent preference from transient context. A recent request may reflect a lasting preference, a temporary task constraint, or noise. SPRInG treats this as a drift-selection problem before updating a user adapter [Kim and Kim, 2026]. In an agent, the distinction is harder because failure may come from task reasoning, tool execution, or a wrong preference assumption. A useful update must identify which part of the trace is evidence about the user.

P3: handle cold start without losing ownership. Sharing population structure improves few-shot adaptation, as explored by collaborative, group-level, and federated personalized models [Choi et al., 2025, Zhang et al., 2025, Zhu et al., 2026]. Yet unrestricted cross-user transfer conflicts with the one-user ownership boundary. The problem is to use a consented prior while keeping the target user's updates, state, and evaluation distinct.

2.2 Moving from offline history to online learning

P4: build an offline initializer with online headroom. The offline checkpoint must already complete tasks, otherwise online interaction produces little useful feedback. It must also remain plastic. OPPU supplies a practical per-user initializer [Tan et al., 2024]; WebRL shows why a warm start matters for sparse-reward agent learning [Qi et al., 2025]. High offline fit alone is not enough if the resulting policy cannot improve from new evidence.

P5: cross the offline–online boundary without a performance dip. Offline records were produced by older policies and may not cover actions preferred by the current policy. The LLM literature studies policy-generator synchronization [Lanchantin et al., 2025]; hybrid preference learning uses offline pairs to reduce online queries [Bose et al., 2024]; and policy expansion protects useful offline behavior during exploration [Zhang et al., 2023]. Our version must additionally preserve user identity, feedback provenance, and checkpoint lineage.

P6: adapt to drift without forgetting or chasing noise. Continual personalization must learn changed preferences while retaining stable ones. SPRInG selects high-drift interactions and keeps residual memory [Kim and Kim, 2026]. Asynchronous LLM training adds another source of staleness because behavior and learner policies diverge [Huang et al., 2026]. Success therefore requires both preference-level retention and valid use of old trajectories.

P7: learn quickly while minimizing user burden. Online feedback is expensive. The method should choose interactions that distinguish plausible user preferences, rather than requesting labels for easy or redundant examples. Hybrid preference optimization formalizes the offline-data versus online-query tradeoff [Bose et al., 2024]; FABLE treats per-user execution choices as an online bandit [Jin et al., 2026]. In our setting, quick adaptation means fewer real interactions or corrections to reach a fixed future-task utility, not merely fewer optimizer steps.

2.3 Learning an agent policy

P8: preserve task completion while personalizing execution. Personalized agents must choose tools, memory, clarifications, action order, and response style while still finishing the task. PersonalWAB evaluates personalized web actions [Cai et al., 2025]; FingerTip 20K separates proactive suggestions from personalized mobile execution [Yang et al., 2026]; and FABLE explicitly separates execution choices from host-agent reasoning [Jin et al., 2026]. These works expose why response preference alone is insufficient: the same final answer may hide a bad or unwanted trajectory.

P9: assign credit from long, partially observed trajectories. The user may provide one correction after many reasoning and tool steps. Sparse task outcomes say whether the task ended well, but not which step expressed the user's preference. WebRL addresses sparse terminal feedback and policy drift for web agents [Qi et al., 2025]; our problem adds user-conditioned credit and delayed preference evidence.

2.4 Obtaining credible and affordable evidence

P10: use simulators without training to simulator artifacts. UserRL, UserVille, ProPerSim, and CollabLLM use simulated interaction to train or evaluate user-facing agents [Qian et al., 2025, Sun et al., 2025, Kim et al., 2025, Wu et al., 2025]. PersonalLLM also creates heterogeneous synthetic preference models [Zollo et al., 2024]. Simulation makes online learning affordable, but it can encode the simulator's style and reward biases. Reward-model accuracy itself may fail to predict downstream personalization quality [Rezk et al., 2025]. A simulator can generate training experience; it cannot be the sole judge of a real-user claim.

P11: make per-user deployment cheaper than repeated large-model use. Dedicated adapters create storage and serving overhead, while full online training adds accelerator cost. OPPU motivates lightweight user ownership [Tan et al., 2024]; CoSteer keeps context local while steering a cloud model [Lv et al., 2025]; and FABLE moves adaptation into a small external policy [Jin et al., 2026]. Our stronger efficiency question is whether a 4B, 7B, or 32B user policy can match or exceed a much larger generic model on user-conditioned utility at lower serving cost.

P12: evaluate future behavior rather than history recall. LongLaMP provides user-disjoint and chronological long-form tasks [Kumar et al., 2024]. PrefEval tests explicit and implicit preference following across long conversations [Zhao et al., 2025]. PERMA adds gradual preference formation, temporal interference, and task completion [Liu et al., 2026]; PRISM supplies individualized human feedback [Kirk et al., 2024]. These benchmarks show that retrieval accuracy is not the endpoint. The endpoint is correct behavior on an untouched future task.

Table 1: The problem map. Each row is a separate empirical claim; success on one row does not imply success on another.

Problem	Main failure mode	Required evidence
Sparse preference inference	The model learns population averages or overfits a few labels.	Seen/unseen behavior, few-shot curves, and worst-user results.
Mixed personal history	Conversation text, task outcomes, and corrections are treated as interchangeable supervision.	Record-type ablations and held-out future behavior.
Offline initialization	The policy is competent but not personal, or personal but unable to complete tasks.	Separate offline task and personalization scores.
Offline–online handoff	Old evidence is stale and the first online updates reduce performance.	Transition curves, behavior-policy logging, and matched update budgets.
Drift and forgetting	The policy follows noise or erases earlier stable preferences.	Controlled shifts, backward transfer, recovery time, and false-update rate.
Feedback efficiency	Improvement requires too many labels, edits, or clarifications.	Utility versus real interactions and user time.
Agent credit assignment	A successful task receives the wrong personalized update.	Step or trace attribution ablations and trajectory-level evaluation.
Simulator validity	The policy optimizes simulator artifacts.	Held-out human agreement, calibration, and method-rank transfer.
Small-model efficiency	Personalization gains disappear under matched cost or stronger models.	Quality–cost Pareto curves and matched large-model baselines.
Ownership and privacy	Raw history leaks across users or to an undeclared service.	Data-flow audit, leakage tests, and explicit communication volume.

3 The Claim We Should Make

3.1 Central claim

> **A compact, user-owned language-model policy can be initialized from one person's heterogeneous historical interactions and then improved from that person's later task evidence, with fewer online interactions than online-only learning, while preserving task success and safety.**

This claim joins three pieces that prior work usually separates:

1. **Personal policy initialization.** The offline stage learns a persistent user-conditioned policy, rather than only retrieving a profile.
2. **Real offline-to-online adaptation.** Later evidence causes positive parameter updates and new loadable checkpoints.
3. **Agent validity.** Improvement is measured on both personalized behavior and task completion over untouched future tasks.

The contribution is therefore the *phase-coupling algorithm*: how historical evidence, fresh trajectories, feedback, replay, and checkpoint activation interact for one user. The contribution is not the existence of LoRA, retrieval, a user simulator, or an online optimizer by itself.

3.2 Secondary claims

These claims should appear only when their dedicated tests pass:

Fast adaptation. The method reaches a fixed utility with fewer real interactions than SPRInG-style selective SFT, naive cumulative fine-tuning, and online-only learning.

Trajectory advantage. Tool actions and verified outcomes improve future personalized execution beyond text-only user history.

Small versus large. A compact personalized policy beats a larger generic model on personalization while remaining task-non-inferior and cheaper to serve.

Simulation benefit. Simulator experience lowers real-user sample requirements without changing the ordering of methods under held-out human evaluation.

Privacy by design. The implementation satisfies a declared data-flow boundary and does not expose another user's raw records. Local PEFT alone is not evidence for this claim.

4 Success Criteria

4.1 Hard validity gates

Before comparing scores, every reported update must pass:
1. the parent and candidate checkpoint hashes differ;
2. the candidate loads in a fresh worker and actually generates the evaluated trajectories;
3. offline, online-adaptation, validation, and future-test intervals are chronological and disjoint;
4. every training record has user, task, behavior-policy, checkpoint, timestamp, and feedback provenance;
5. failed runs, users, and rollbacks remain in the report;
6. the simulator, reward model, or LLM judge has not seen the future-test target or its answer.

Failure of any item invalidates the run rather than lowering its score.

4.2 Primary statistical gate

Let b be the strongest eligible baseline on the validation set. For each user-period cluster, evaluate the candidate and b on identical future tasks. Define

$$\Delta_{\text{pers}} = U_{\text{pers}}(c_{\text{candidate}}) - U_{\text{pers}}(c_b), \tag{3}$$

$$\Delta_{\text{task}} = U_{\text{task}}(c_{\text{candidate}}) - U_{\text{task}}(c_b). \tag{4}$$

The main claim passes only when

$$\text{LCB}_{95\%}(\Delta_{\text{pers}}) > 0 \quad \text{and} \quad \text{LCB}_{95\%}(\Delta_{\text{task}}) \geq -\delta_{\text{task}}. \tag{5}$$

The minimum meaningful personalization effect and δ_{task} must be fixed before final evaluation. Confidence intervals are clustered by user or task lineage, not by treating correlated turns as independent samples.

4.3 What "beat a large model" means

The large-model comparison must include:
1. a large generic model without personal history;
2. the same model with the full allowed history;
3. strong RAG or profile prompting;
4. a decoding-time method such as CoSteer when executable [Lv et al., 2025];
5. the compact offline-only and compact online-adapted policies.

Prompts, tools, history access, and task budgets must match. We may claim that the compact model beats the large model only if it improves personalization, meets task non-inferiority, and costs less. A win on preference style alone does not establish superior task capability.

Table 2: Claim-specific success criteria. The first four rows are required for the central claim; later rows authorize narrower secondary claims.

Claim	Measurement	Pass condition
Personalized improvement	Paired future-task Δ_{pers} against the strongest baseline.	Lower 95% bound is above zero and the point estimate exceeds the preregistered meaningful effect.
Task preservation	Paired task success or environment reward.	Lower 95% bound exceeds $-\delta_{\text{task}}$.
Value of online learning	Candidate versus frozen offline checkpoint and naive cumulative update.	Candidate passes equation (5) against both.
Real policy change	Parameter count, hashes, load receipt, and changed outputs.	Nonzero finite update; candidate is loaded for evaluation.
Fast adaptation	Area under the future-utility learning curve and interactions-to-target.	Higher clustered 95% interval for area, or fewer interactions to the same preregistered target.
Retention	Backward transfer on earlier stable preferences and tasks.	No task-margin violation; less forgetting than naive continual training.
Drift recovery	Utility after a controlled persistent preference change.	Faster recovery without elevated updates on transient-noise controls.
Safety	Severe violations and policy regressions under a fixed panel.	No increase in severe events; lower-severity change stays within its preregistered margin.
Small versus large	Compact personalized model versus large generic, RAG, and profile-enabled models under equal information.	Personalization superiority, task non-inferiority, and strictly lower serving cost all hold.
Simulator benefit	Human–simulator agreement, calibration, and method ranking.	Simulator-trained gain survives held-out human evaluation; simulation is never the only final judge.
Efficiency	Training GPU-hours, inference latency, memory, storage, tokens, API cost, and user-feedback count.	Candidate lies on the quality–cost Pareto frontier under matched budgets.
Ownership/privacy	Raw-record flow, communication bytes, cross-user leakage probes.	Declared boundary is met with no unauthorized raw-history transfer.

4.4 What quick adaptation means

Report utility after every fixed number of real interactions, not merely after epochs. For threshold τ,

$$T_\tau = \min\{n : U_{\text{pers}}(c_u^{(n)}) \geq \tau \text{ and task/safety gates pass}\}. \tag{6}$$

Also report area under the learning curve, final utility, number of clarifications, user correction time, and rollbacks. A method is faster only if it reaches the same valid target with less user evidence or wall-clock cost.

4.5 What simulator success means

Simulation has three separate tests:

1. **Preference fidelity:** agreement and calibration on held-out real-user comparisons.
2. **Method fidelity:** rank correlation between methods under the simulator and under real-user evaluation.

Table 3: Minimum baseline families.

Family	Purpose	Representative implementation
No personalization	Measures the value of any user information.	Same-size base and large API model.
Prompt/profile/RAG	Tests whether parameter updates are necessary.	Full-history prompting, retrieval, and generated profile.
Offline per-user PEFT	Tests the value of the online phase.	OPPU [Tan et al., 2024].
Shared personalized policy	Tests one-user ownership against a population model.	P-RLHF, CoPL, or PROPER [Li et al., 2024, Choi et al., 2025, Zhang et al., 2025].
Continual per-user update	Tests the proposed transition and replay rule.	SPRInG and naive cumulative SFT [Kim and Kim, 2026].
Online-only	Measures the value of offline initialization.	Same optimizer initialized from the generic checkpoint.
Offline then naive online	Isolates the new coupling mechanism.	Sequential SFT/DPO/RL with no special replay or activation rule.
Weight-free online control	Tests whether policy training is worth its cost.	FABLE and CoSteer [Jin et al., 2026, Lv et al., 2025].
Large-model teacher	Tests compact-model competitiveness.	The CLI/API model that produced the offline trajectories.

3. **Transfer value:** fewer real interactions are needed after simulator pretraining than without it.

Passing only simulator reward permits a simulator result, not a real-user personalization claim.

5 Experimental Comparison Set

Evaluation substrates. Use at least one chronological generation stream such as LongLaMP [Kumar et al., 2024] and one executable agent substrate such as PersonalWAB or FingerTip 20K [Cai et al., 2025, Yang et al., 2026]. PrefEval and PERMA provide additional long-horizon preference tests [Zhao et al., 2025, Liu et al., 2026]. User simulators can support training, but the final claim needs held-out human feedback or frozen real-user records. A result on one substrate is a feasibility result; a general claim requires at least two materially different substrates.

Replication. Although each deployed policy belongs to one user, a paper should repeat the same protocol over many independent users or user streams. Report per-user results, the median, the worst decile, and cluster-aware intervals. Use at least three seeds for feasibility and five for final claims.

6 Claims We Must Not Make

- Chronological offline replay is not post-deployment online learning.
- Better retrieval or profile recall is not evidence of a better policy.
- A simulator-only win is not evidence of real-user improvement.
- A changed loss value is not evidence that the candidate checkpoint was loaded.
- A personalized reward-model gain is not automatically a policy or task gain [Rezk et al., 2025].
- Local adapters do not by themselves prove privacy.
- Beating a large model on style does not mean beating it on task capability.
- Mean improvement cannot hide severe failures for a subset of users.

7 Recommended Paper Position

The strongest paper is not a survey of every personalization problem. It is an algorithm paper about the transition from *one user's historical evidence* to *one user's continually improving policy.* Data adaptation defines the offline input. Offline-to-online optimization is the method. Task-preserving future personalization is the primary result. Quick adaptation, simulator-assisted training, privacy, and small-model efficiency are valuable secondary claims, each earned by its own test.

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